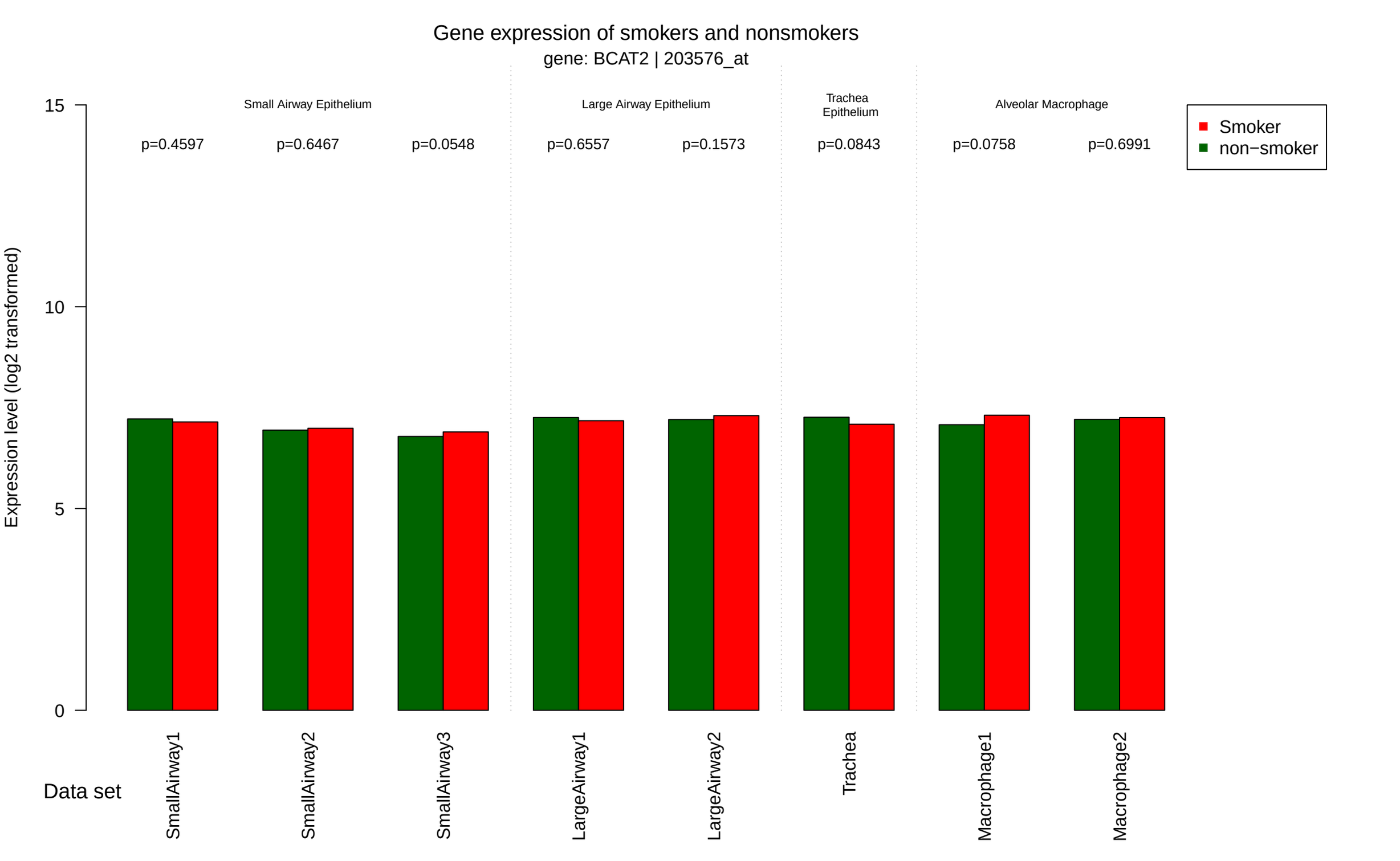
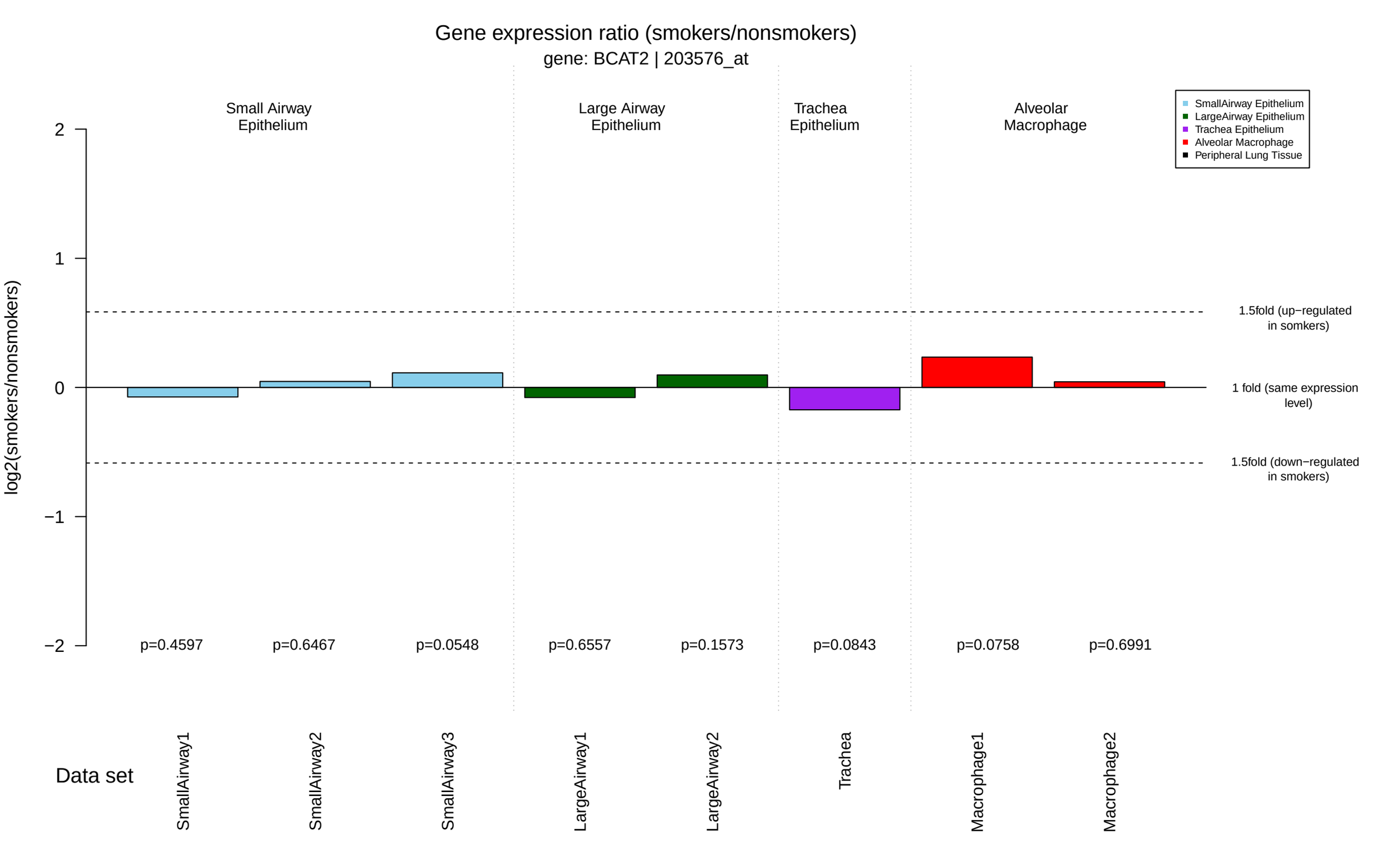
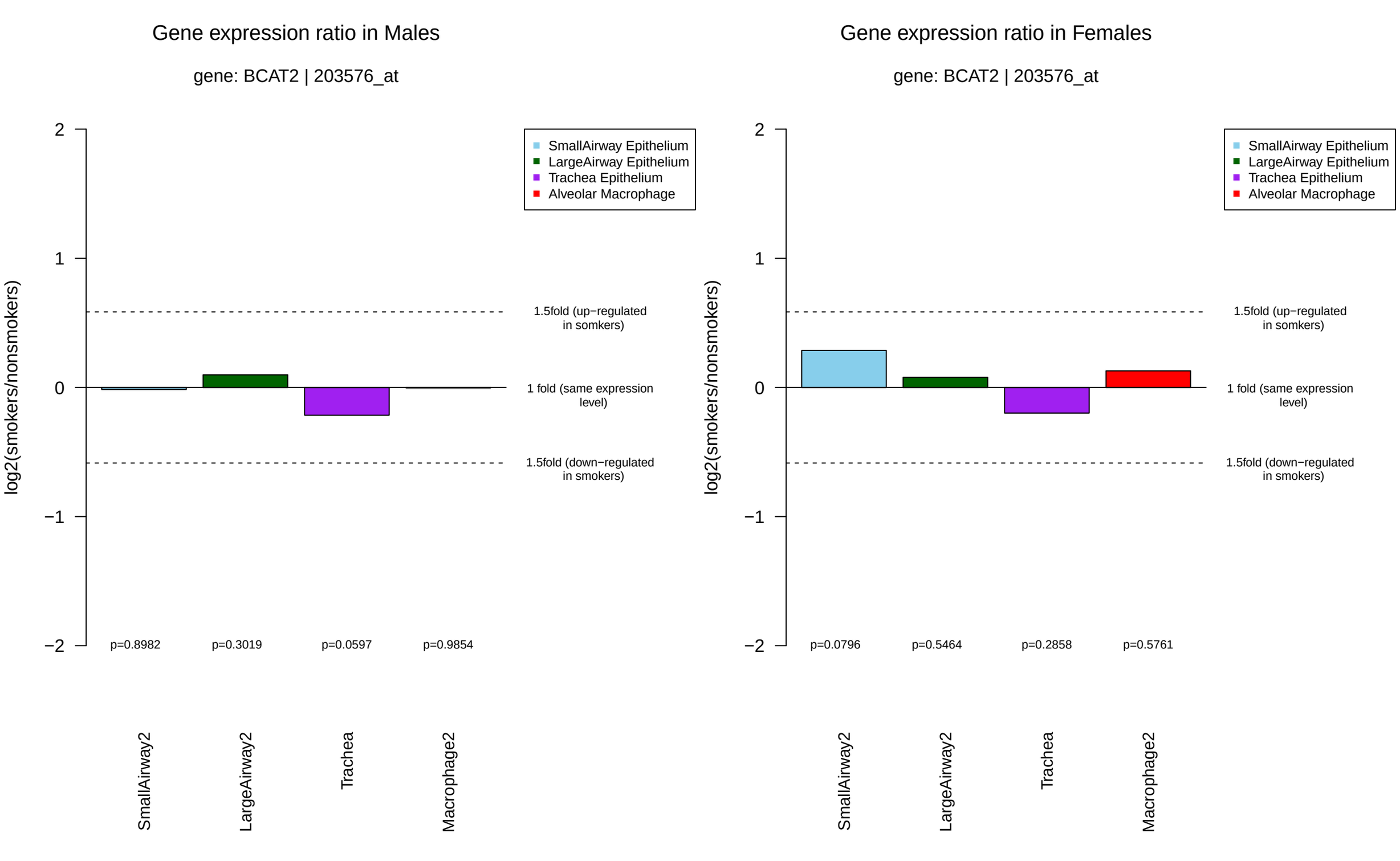
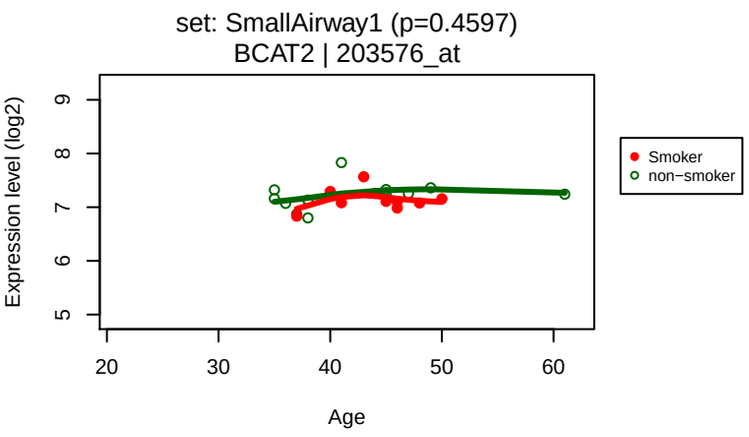
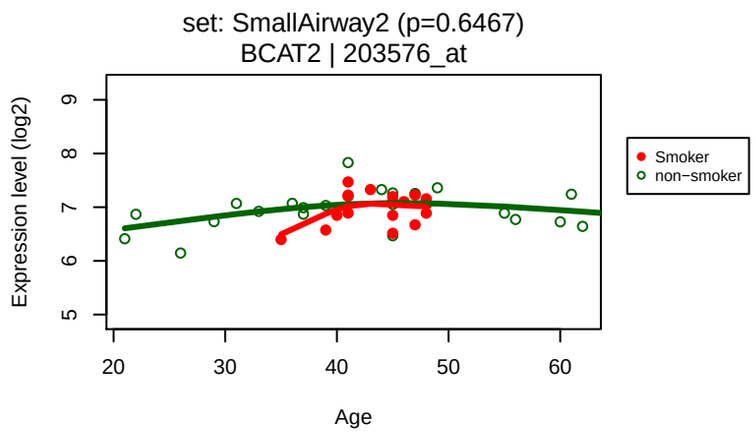
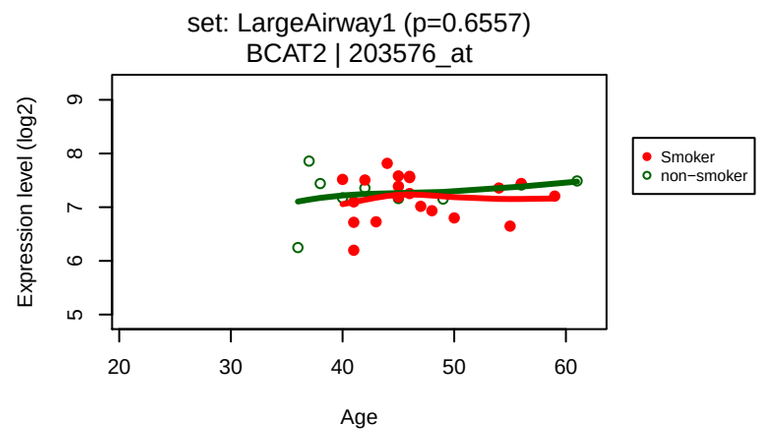

Gene expression of smokers and nonsmokers
gene: BCAT2 | 203576_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4597
p=0.6467
p=0.0548
p=0.6557
p=0.1573
p=0.0843
p=0.0758
p=0.6991
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

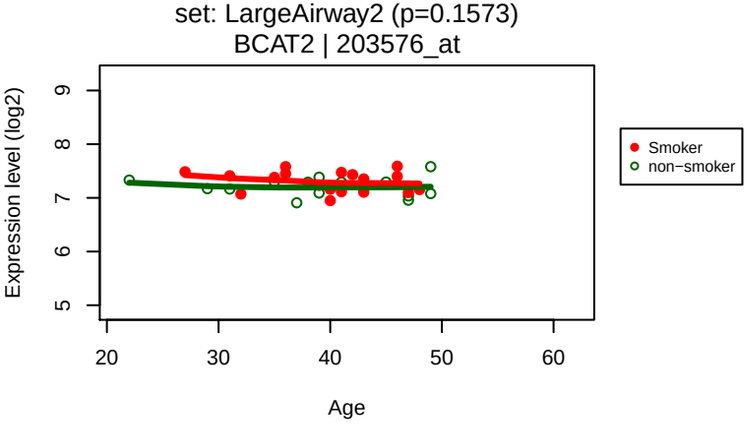

Gene expression ratio (smokers/nonsmokers)
gene: BCAT2 | 203576_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4597
p=0.6467
p=0.0548
p=0.6557
p=0.1573
p=0.0843
p=0.0758
p=0.6991
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

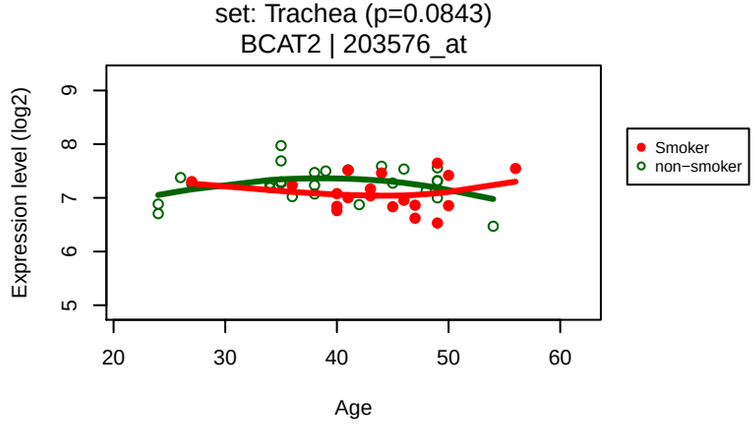

Gene expression ratio in Males
gene: BCAT2 | 203576_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8982
p=0.3019
p=0.0597
p=0.9854
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BCAT2 | 203576_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0796
p=0.5464
p=0.2858
p=0.5761
SmallAirway2
LargeAirway2
Trachea
Macrophage2

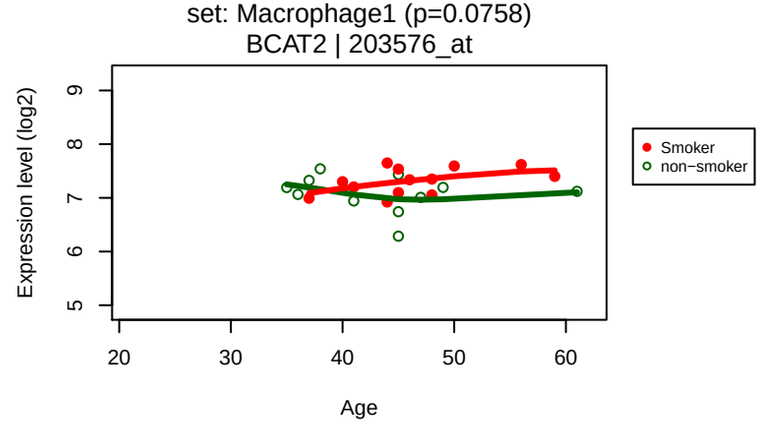

set: SmallAirway1 (p=0.4597)
BCAT2 | 203576_at
Expression level (log2)
Age
Smoker
non−smoker

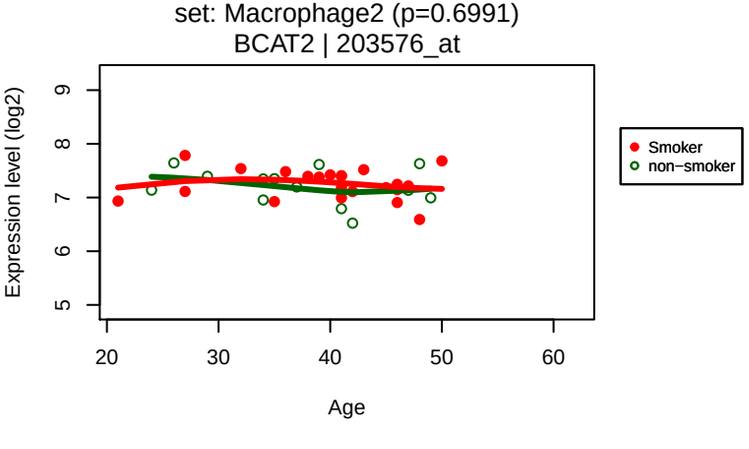

set: SmallAirway2 (p=0.6467)
BCAT2 | 203576_at
Expression level (log2)
Age
Smoker
non−smoker

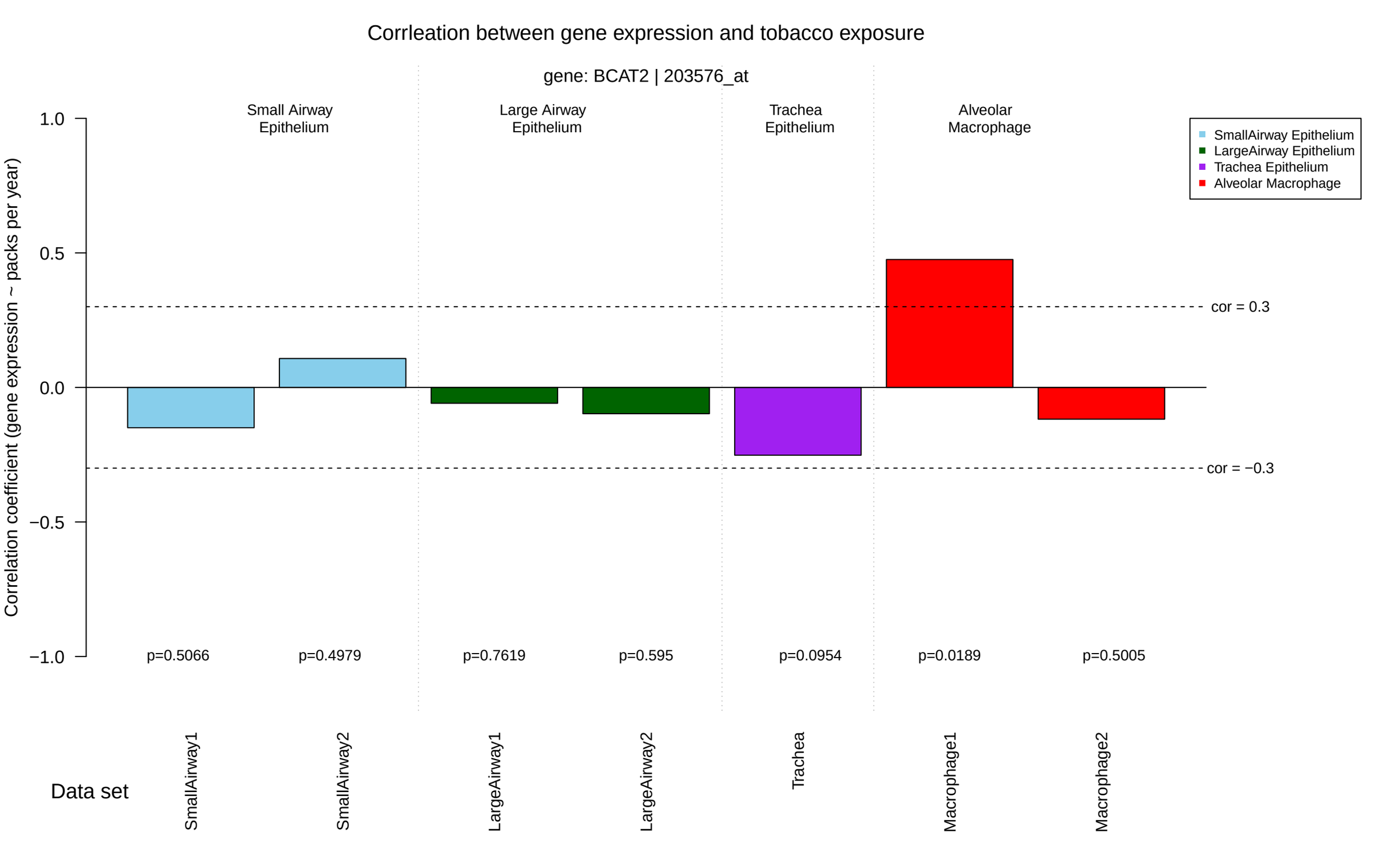

set: LargeAirway1 (p=0.6557)
BCAT2 | 203576_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1573)
BCAT2 | 203576_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0843)
BCAT2 | 203576_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0758)
BCAT2 | 203576_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6991)
BCAT2 | 203576_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BCAT2 | 203576_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5066
p=0.4979
p=0.7619
p=0.595
p=0.0954
p=0.0189
p=0.5005
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2